Network Layer: Internet Protocol

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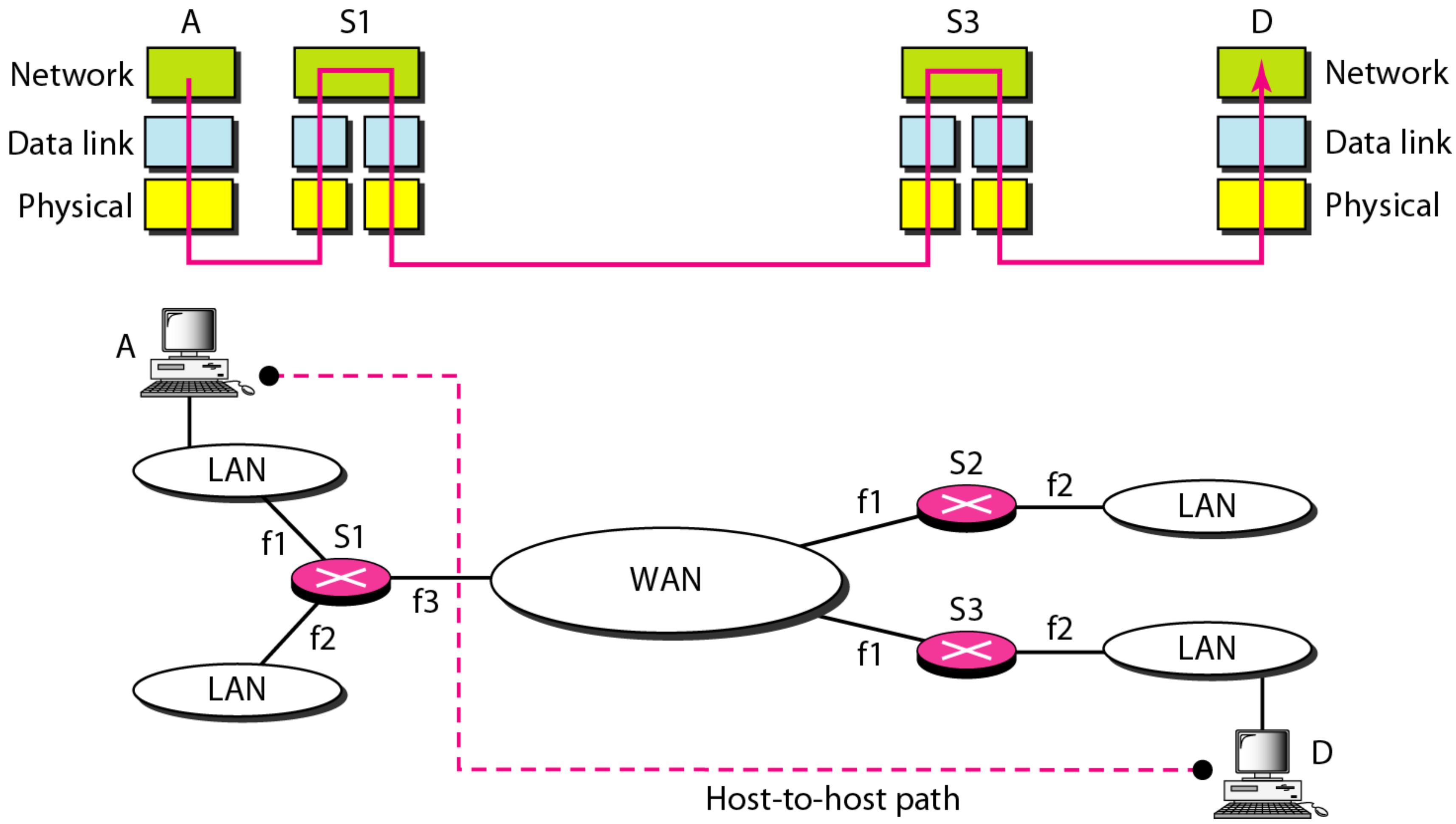
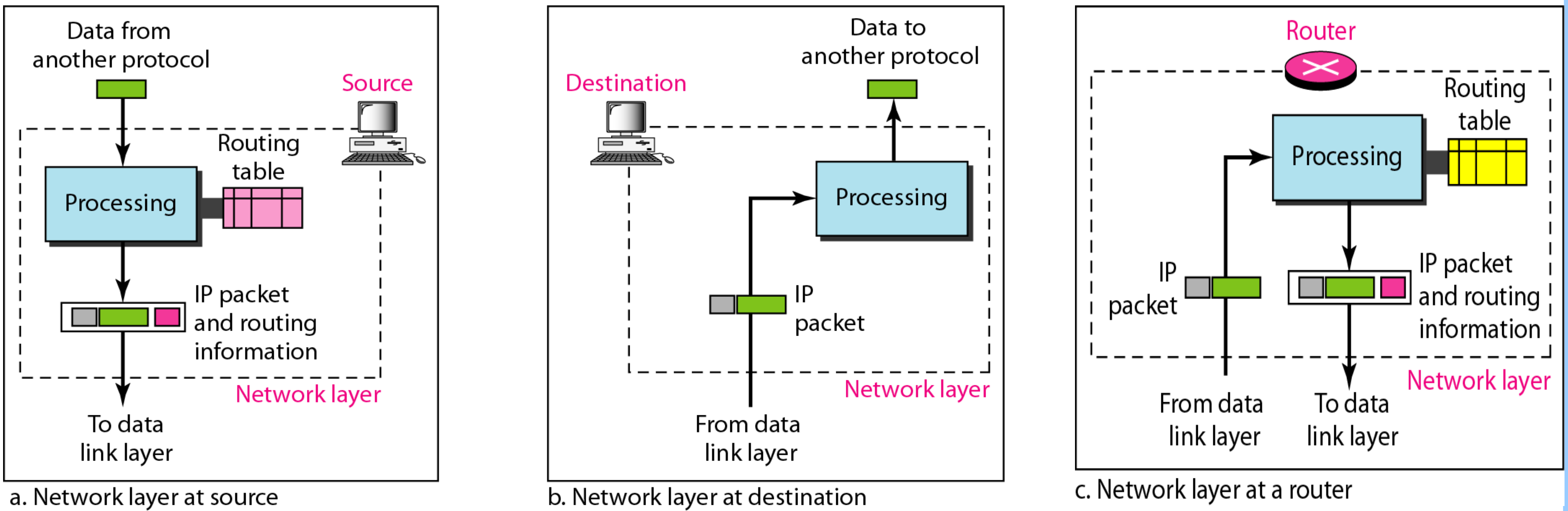
Need of network layer



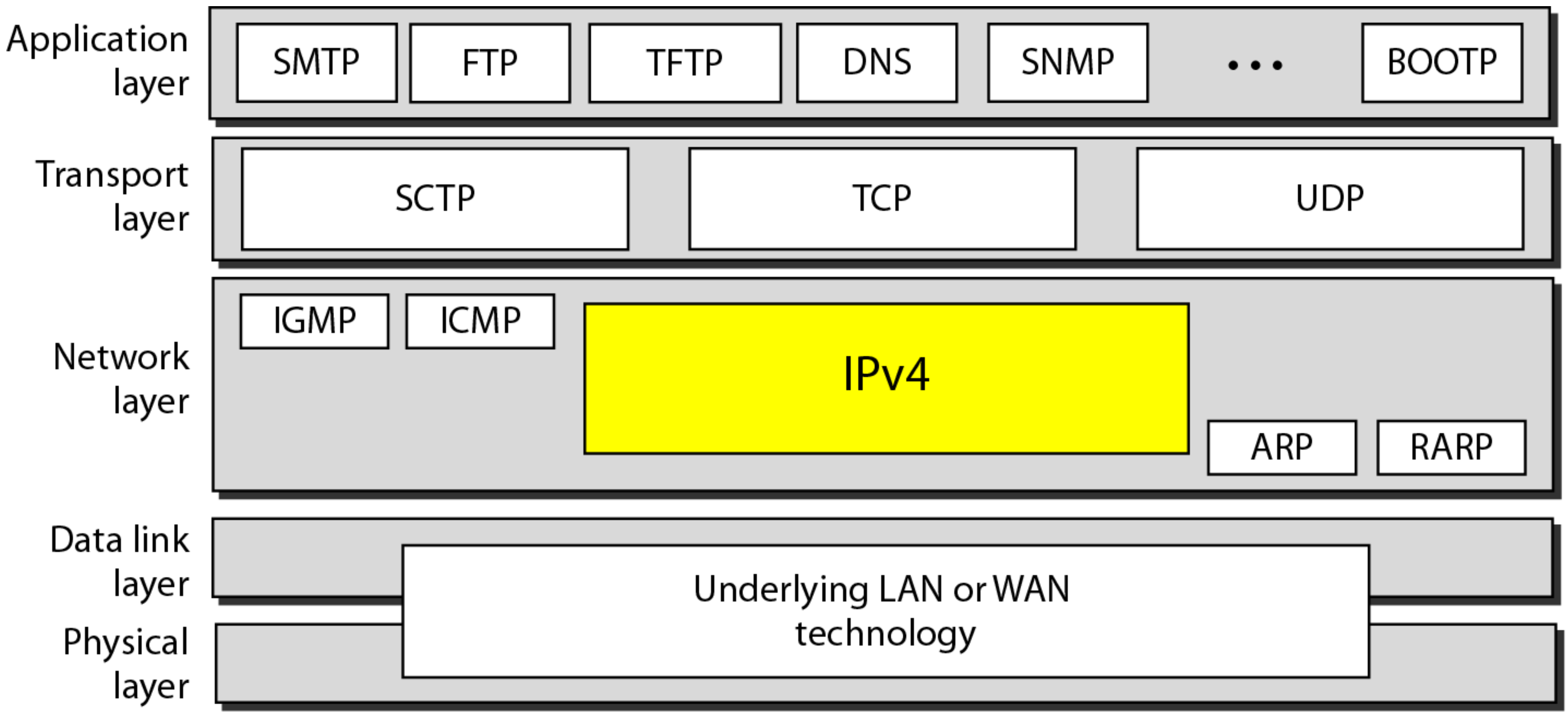
The task of Network Layer



The Internet Protocol version 4 (IPv4)

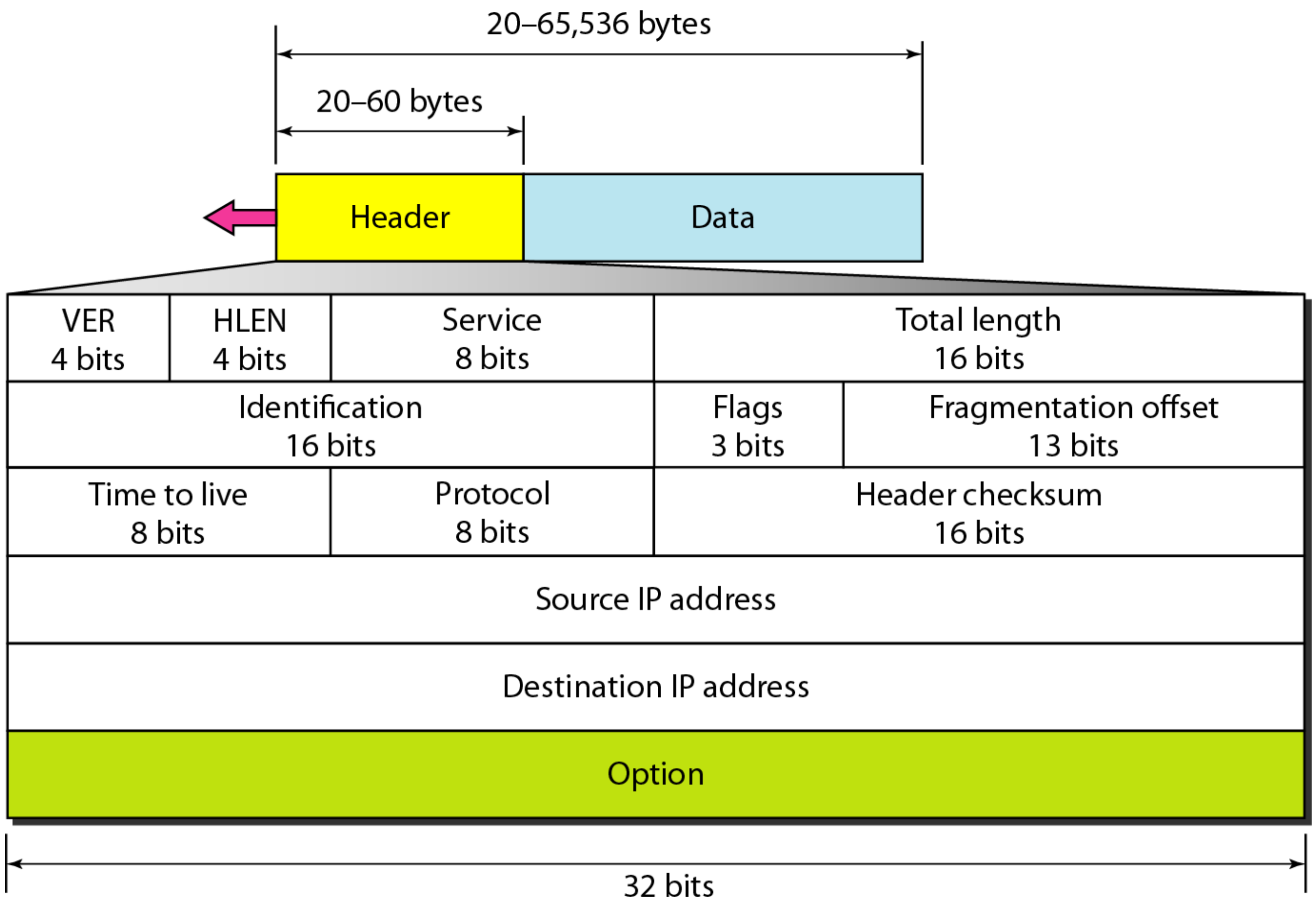
- IPv4 is an unreliable and connectionless datagram protocol-a best-effort delivery service
- What is mean by best-effort service?
  - IPv4 provides no error control or flow control (except for error detection on the header). IPv4 assumes the unreliability of the underlying layers and does its best to get a transmission through to its destination, but with no guarantees.
- If reliability is important, IPv4 must be paired with a reliable protocol such as TCP
- Each datagram is handled independently, and each datagram can follow a different route to the destination
- This implies that datagrams could arrive out of order
- Also, some could be lost or corrupted during transmission

The Internet Protocol version 4 (IPv4)



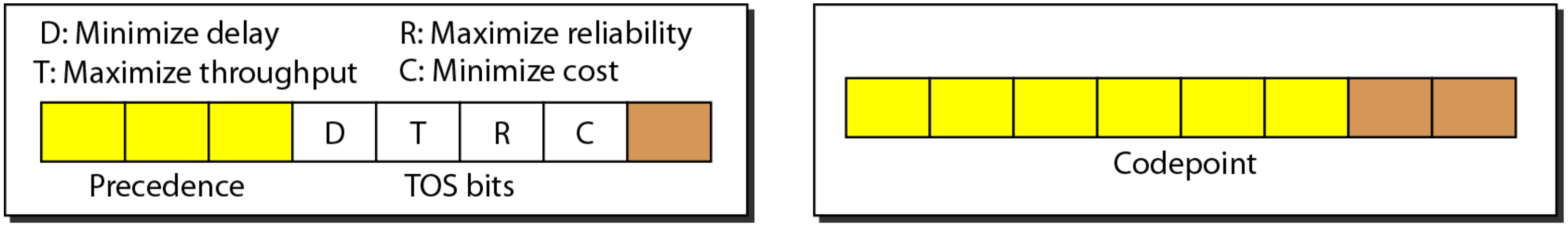
IPv4 Datagram

- Packets in the IPv4 layer are called datagrams



IPv4 Datagram format

- Version (VER)- This 4-bit field defines the version of the IPv4 protocol
- Header length (HLEN)- This 4-bit field defines the total length of the datagram header in 4-byte words
- Services- the first 3 bits are called precedence bits, The next 4 bits are called type of service (TOS) bits, and the last bit is not used



Service type

Differentiated services

IPv4 Datagram format

- Total length : defines the total length (header plus data) of the IPv4 datagram in bytes
- Identification: This 16-bit field identifies a datagram originating from the source host.
- Flags: This is a 3-bit field
  - The first bit is reserved
  - The second bit is called the do not fragment bit
    - If its value is 1, the machine must not fragment the datagram
    - its value is 0, the datagram can be fragmented if necessary
  - The third bit is called the more fragment bit
    - If its value is 1, it means the datagram is not the last fragment
    - If its value is 0, it means this is the last or only fragment

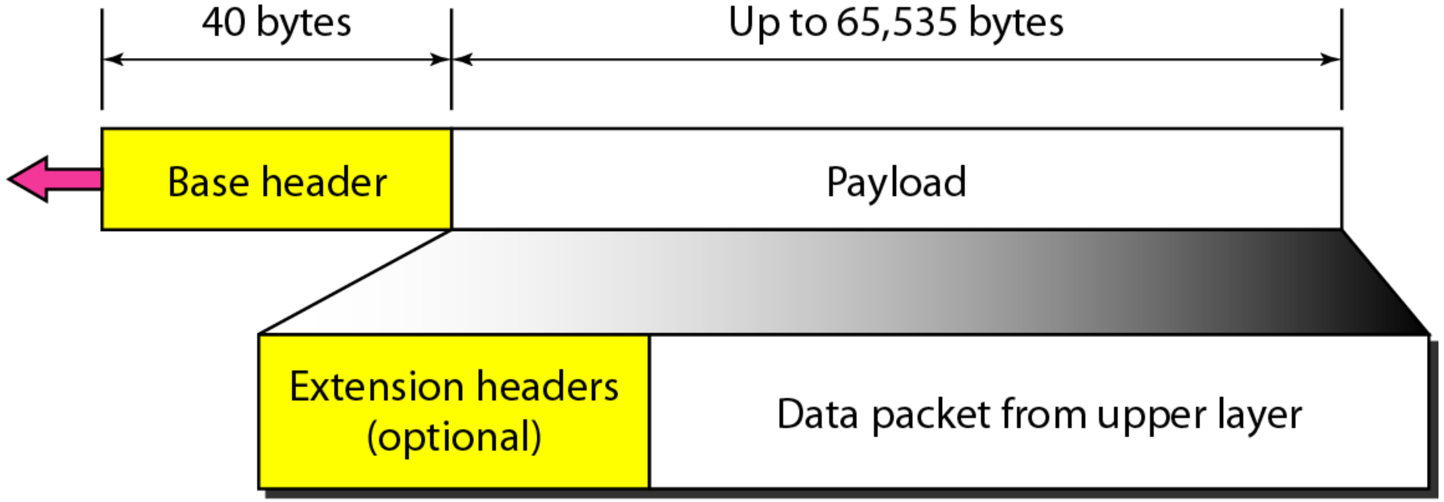
IPv4 Datagram format

- Fragmentation offset: This 13-bit field shows the relative position of this fragment with respect to the whole datagram
- Time to live(TTL):
  - A datagram has a limited lifetime in its travel through an internet
  - Field is used mostly to control the maximum number of hops (routers) visited by the datagram
  - Sender stores a number approximately 2 times the maximum number of routes between any two hosts
  - Each router that processes the datagram decrements this number by 1
- Protocol. This 8-bit field defines the higher-level protocol
  - 1-ICMP, 2-IGMP, 6-TCP, 17-UDP, 89-OSPF

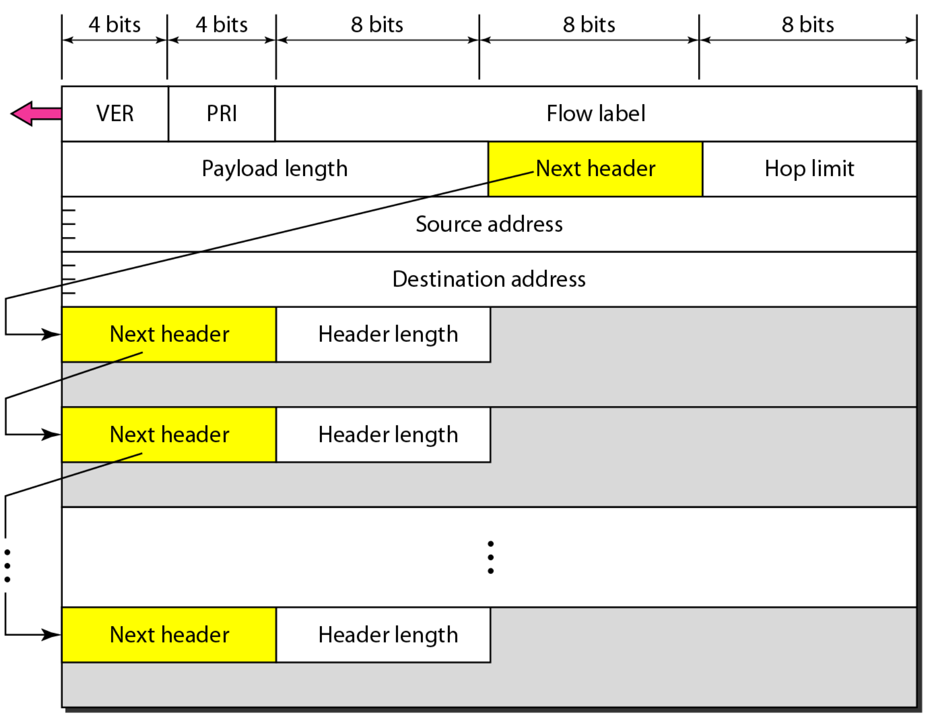
IPv4 Datagram format

- Checksum. The checksum value
- Source address. This 32-bit field defines the IPv4 address of the source
- Destination address. This 32-bit field defines the IPv4 address of the destination
- MTU (Maximum Transfer Unit)
  - field defines the maximum size of the data field
  - When a datagram is encapsulated in a frame, the total size of the datagram must be less than this maximum size
  - Depends on physical topology
- Options, as the name implies, are not required for a datagram. They can be used for network testing and debugging

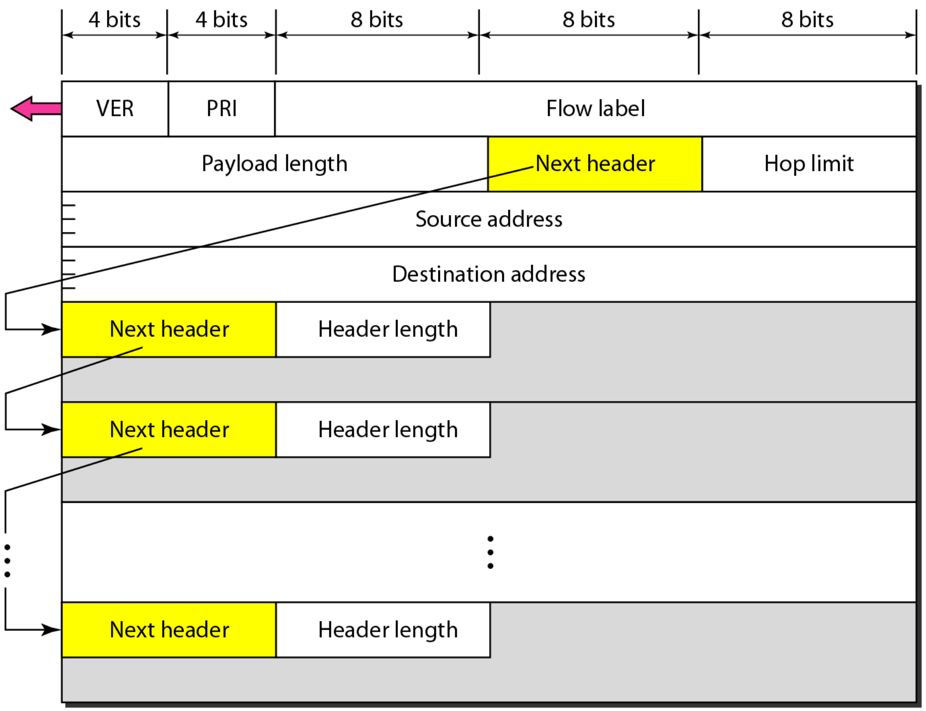
IPv6 Header



- Each packet is composed of a mandatory base header followed by the payload
- The payload consists of two parts: optional extension headers and data from an upper layer
- The base header occupies 40 bytes, whereas the extension headers and data from the upper layer contain up to 65,535 bytes of information.

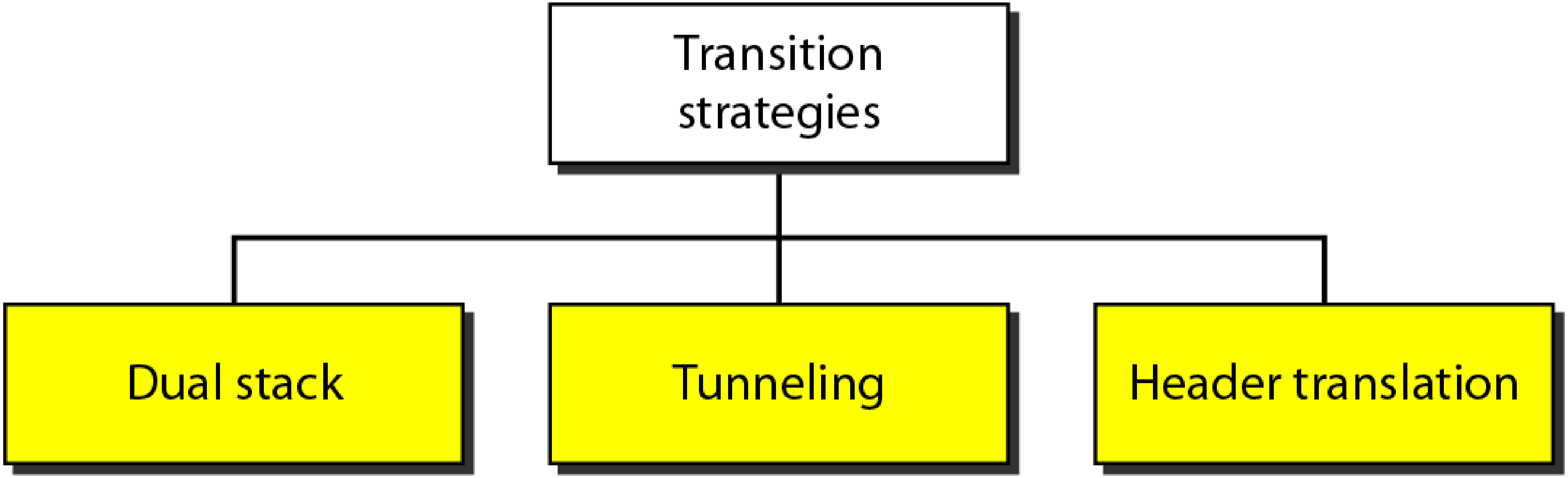


- The base header has eight fields.
  - Version
  - Priority - defines the priority of the packet with respect to traffic congestion
  - Flow label. The flow label is a 3-byte (24-bit) field that is designed to provide special handling for a particular flow of data
  - Payload length. The 2-byte payload length field defines the length of the IP datagram excluding the base header
  - Next header. The next header is an 8-bit field defining the header that follows the base header in the datagram
  - The next header is either one of the optional extension headers used by IP or the header of an encapsulated packet such as UDP or TCP



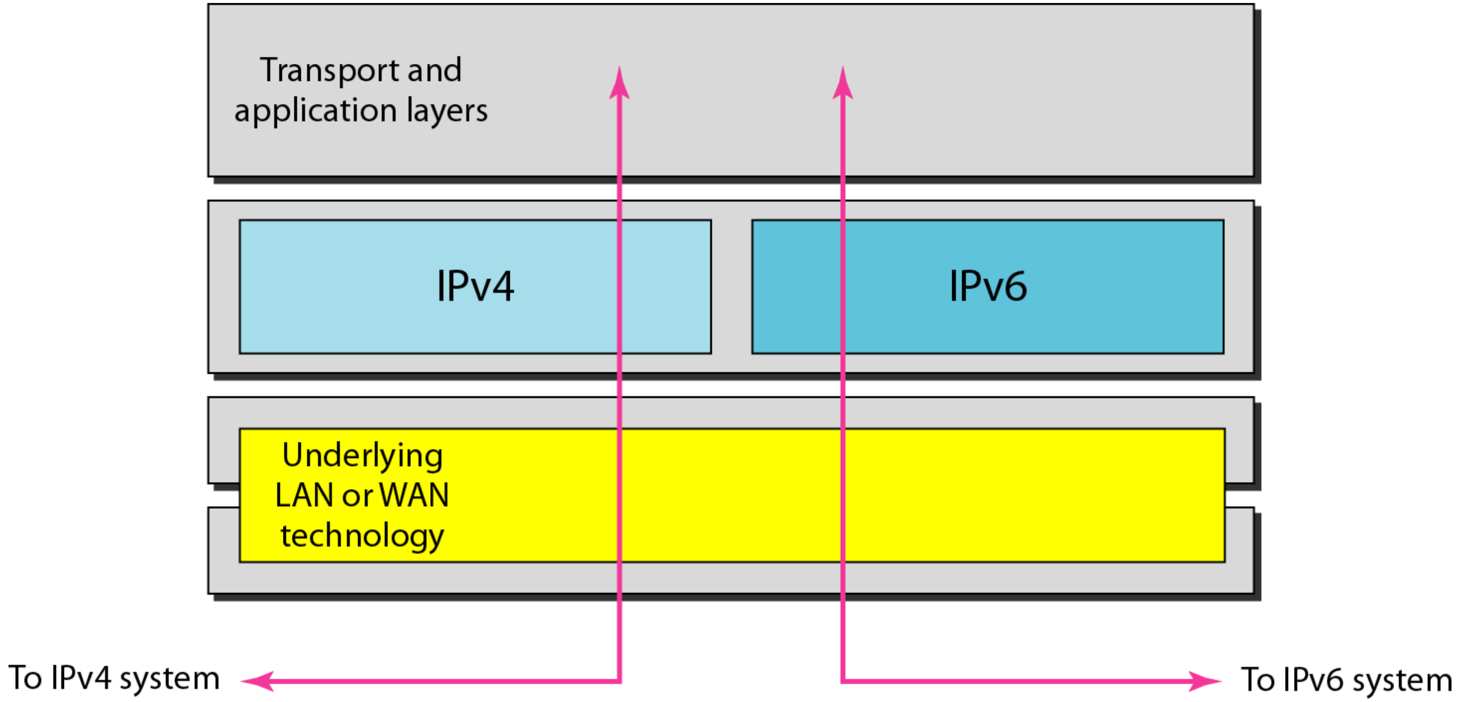
- Hop limit - 8-bit hop limit field serves the same purpose as the TTL field in IPv4.
- Source address - The source address field is a 16-byte (128-bit) Internet address that identifies the original source of the datagram.
- Destination address - The destination address field is a 16-byte (128-bit) Internet address that usually identifies the final destination of the datagram
- However, if source routing is used, this field contains the address of the next router

TRANSITION FROM IPv4 TO IPv6



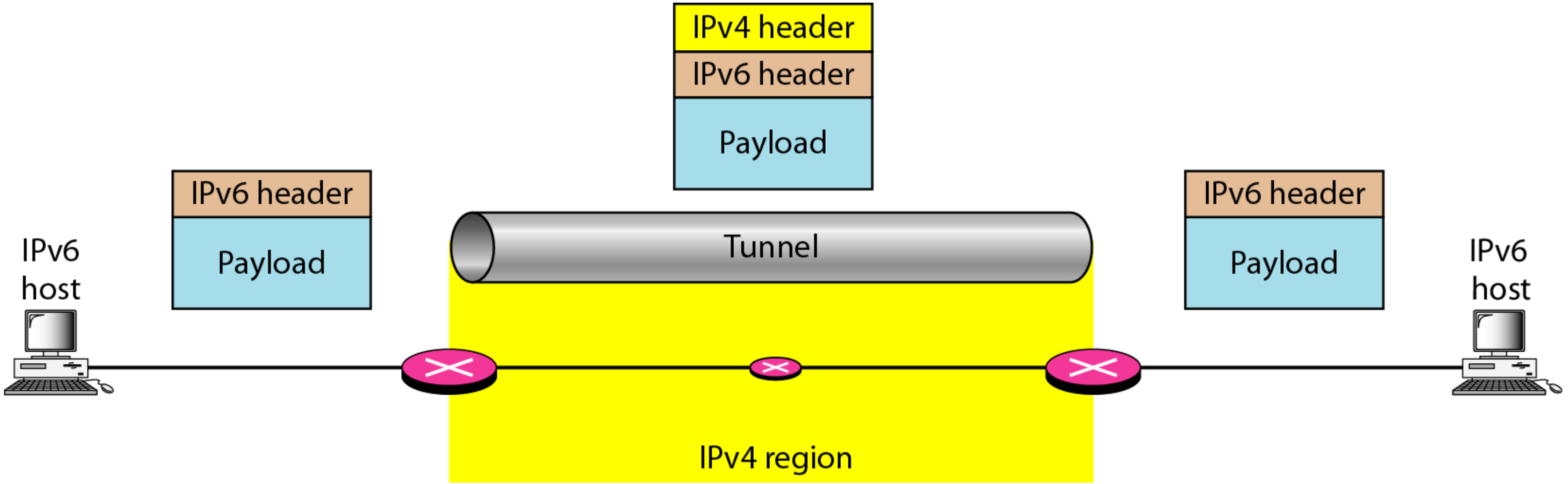
Dual Stack

- Station must run IPv4 and IPv6 simultaneously until all the Internet uses IPv6
- To determine which version to use when sending a packet to a destination, the source host queries the DNS.



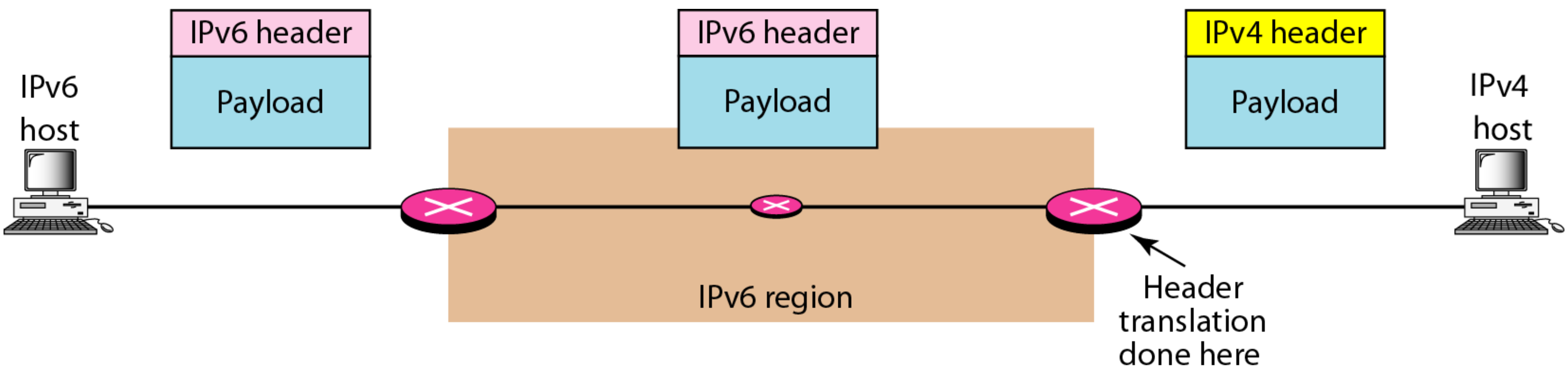
Tunneling

- Tunneling is a strategy used when two computers using IPv6 want to communicate with each other and the packet must pass through a region that uses Ipv4
- The IPv6 packet is encapsulated in an IPv4 packet when it enters the region, and it leaves its capsule when it exits the region
- The IPv4 packet is carrying an IPv6 packet as data, the protocol value is set to 41



Header translation

- Header translation is necessary when the majority of the Internet has moved to IPv6 but some systems still use IPv4
- Mostly the receiver does not understand IPv6
- The header format must be totally changed through header translation. The header of the IPv6 packet is converted to an IPv4 header

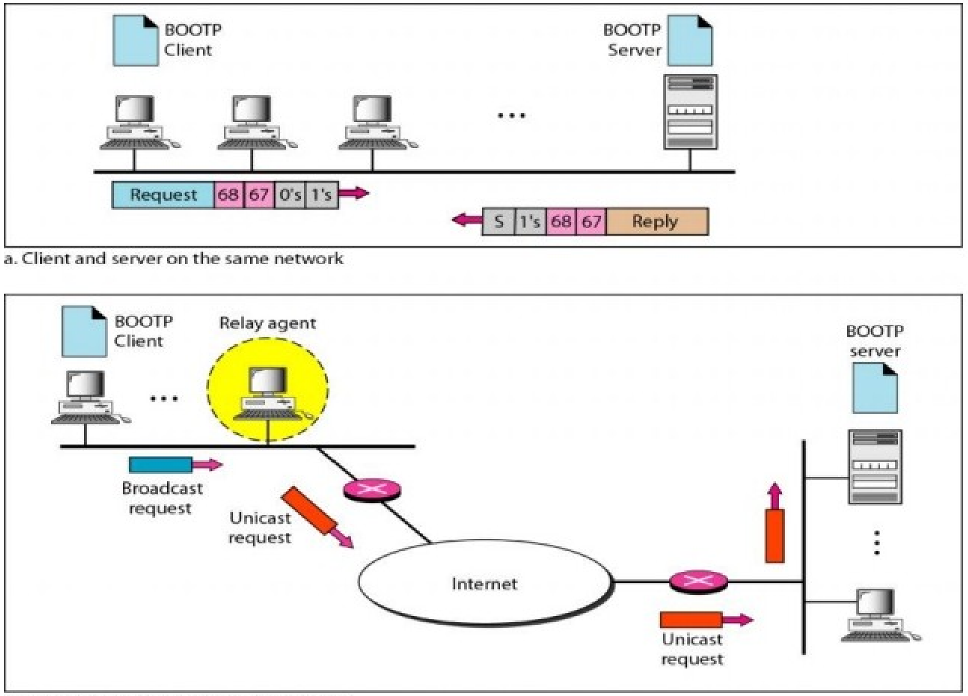


RARP

- Reverse Address Resolution Protocol (RARP) finds the logical address for a machine that knows only its physical address
- A RARP request is created and broadcast on the local network, the machine on the local network that knows all the IP addresses will respond with a RARP reply (client-server approach)
- Broadcasting is done at the data link layer
- An administrator having several networks or several subnets needs to assign a RARP server for each network or subnet
  - This is the reason that RARP is almost obsolete

BOOTP

- The Bootstrap Protocol (BOOTP) is a client/server protocol designed to provide physical address to logical address mapping
- BOOTP is an application layer protocol
- The administrator may put the client and the server on the same network or on different networks



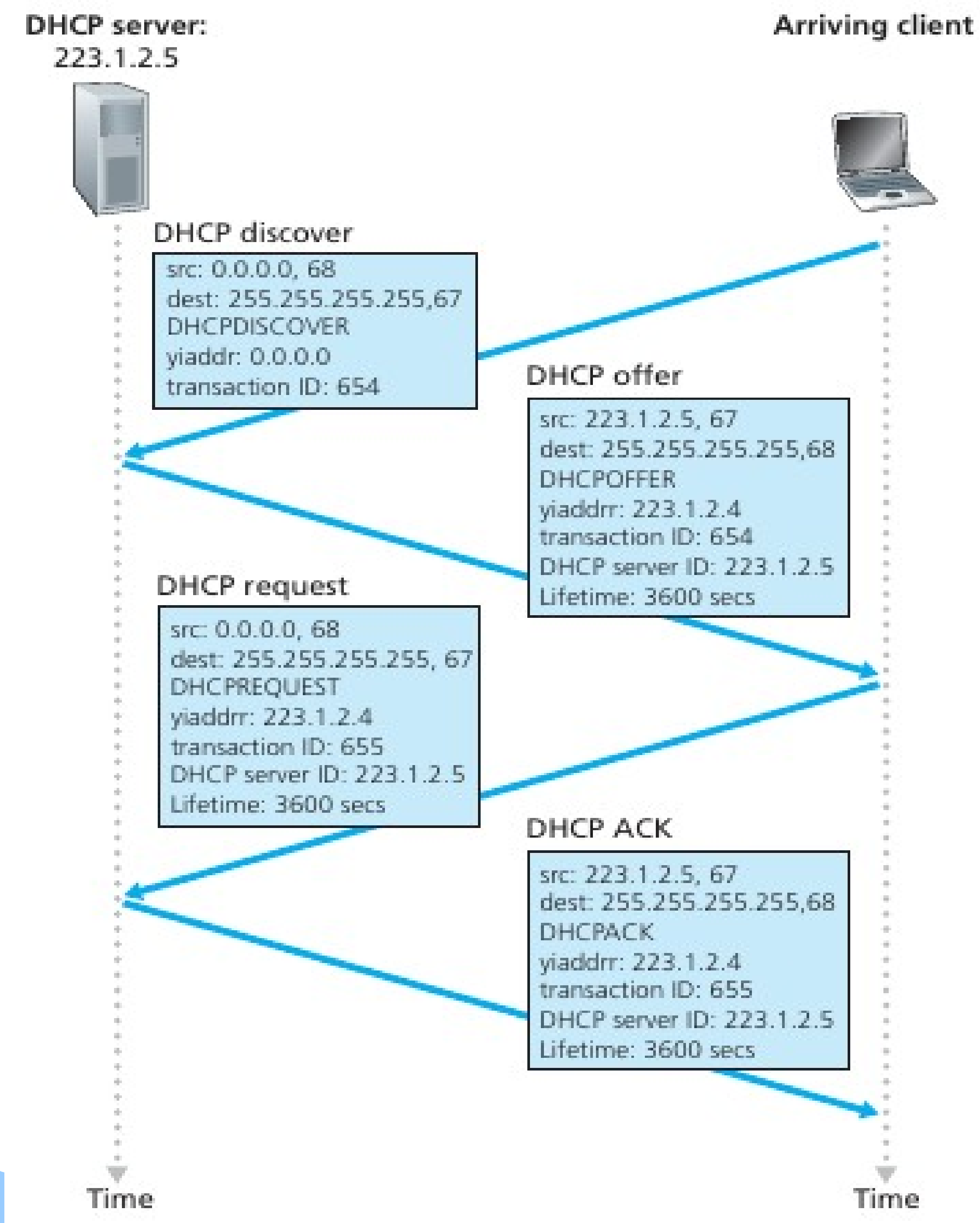
a. Client and server on the same network

b. Client and server on different networks

DHCP

- The Dynamic Host Configuration Protocol (DHCP) has been devised to provide static and dynamic address allocation
- Backward-compatible with BOOTP to provide static address allocation using static address pool bound with physical address
- DHCP has a second database with a pool of available IP addresses
- When a DHCP client requests a temporary IP address, the DHCP server goes to the pool of available (unused) IP addresses and assigns an IP address for a negotiable period of time

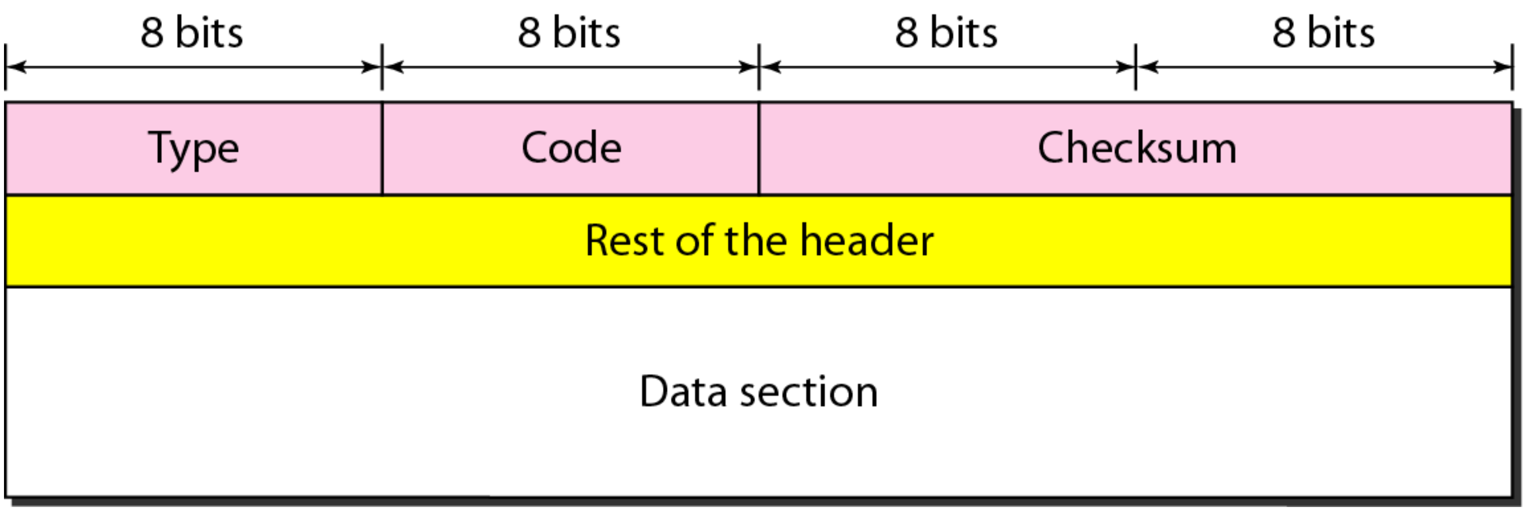
DHCP


DHCP server:
223.1.2.5
Arriving client
DHCP discover
src: 0.0.0.0, 68
dest: 255.255.255.255,67
DHCPDISCOVER
yiaddr: 0.0.0.0
transaction ID: 654
DHCP offer
src: 223.1.2.5, 67
dest: 255.255.255.255,68
DHCPOFFER
yiaddrr: 223.1.2.4
transaction ID: 654
DHCP server ID: 223.1.2.5
Lifetime: 3600 secs
DHCP request
src: 0.0.0.0, 68
dest: 255.255.255.255, 67
DHCPREQUEST
yiaddrr: 223.1.2.4
transaction ID: 655
DHCP server ID: 223.1.2.5
Lifetime: 3600 secs
DHCP ACK
src: 223.1.2.5, 67
dest: 255.255.255.255,68
DHCPACK
yiaddrr: 223.1.2.4
transaction ID: 655
DHCP server ID: 223.1.2.5
Lifetime: 3600 secs
Time
Time

ICMP

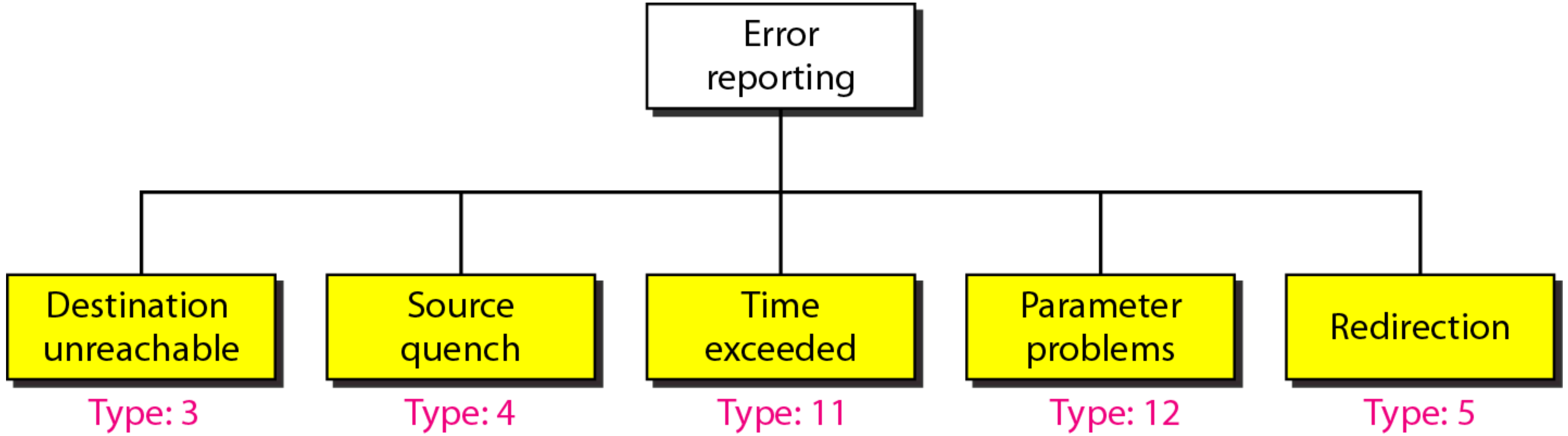
- The IP protocol has no error-reporting or error-correcting mechanism
- The IP protocol also lacks a mechanism for host and management queries
- The Internet Control Message Protocol (ICMP) has been designed to compensate for the above two deficiencies
- It is a companion to the IP protocol
- ICMP messages are divided into two broad categories
  - Error-reporting messages
  - Query messages

Message format



Error Reporting

- One of the main responsibilities of ICMP is to report errors
- Error messages are always sent to the original source
- Five types of errors are handled



Error Reporting

- No ICMP error message will be generated in response
  - To a datagram carrying an ICMP error message
  - For a fragmented datagram that is not the first fragment.
  - For a datagram having a multicast address.
  - For a datagram having a special address such as 127.0.0.0 or 0.0.0.0

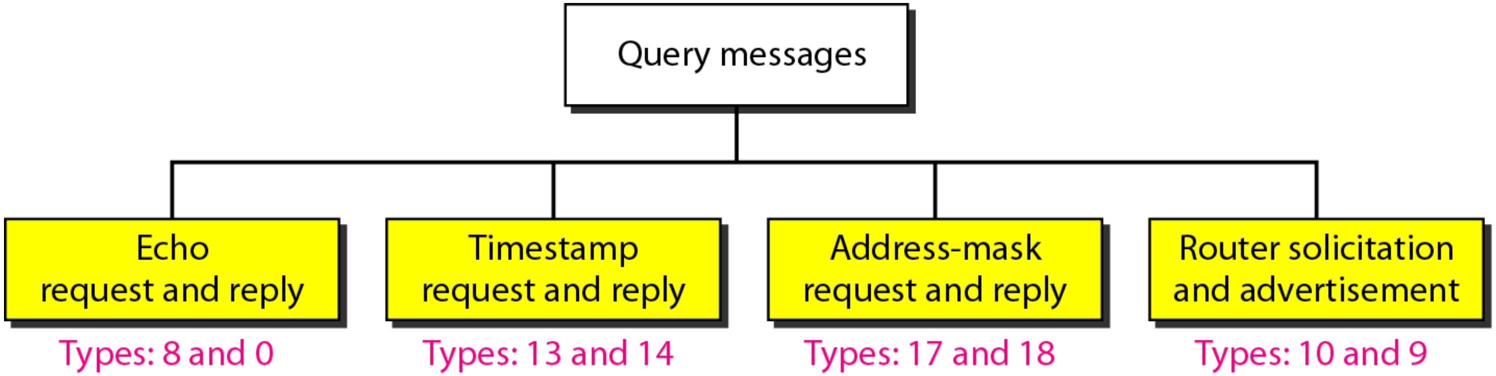
Errors

- Destination Unreachable
  - When a router cannot route a datagram or a host cannot deliver a datagram, the datagram is discarded and the router or the host sends a destination-unreachable message back to the source host that initiated the datagram
- Source Quench
  - A kind of flow control to the IP
  - When a router or host discards a datagram due to congestion, it sends a source-quench message to the sender of the datagram
    - First, it informs the source that the datagram has been discarded
    - Second, it warns the source that there is congestion somewhere in the path
- Parameter Problem
  - If a router or the destination host discovers an ambiguous or missing value in any field of the datagram, it discards the datagram and sends a parameter-problem message back to the source

Errors

- Time Exceeded
  - When the time-to-live value reaches 0, after decrementing, the router discards the datagram and a time-exceeded message must be sent by the router to the original source
  - A time-exceeded message is also generated when not all fragments that make up a message arrive at the destination host within a certain time limit.
- Redirection
  - Due to static routing table, the host may sometime send a datagram to the wrong router. In this case, the router that receives the datagram will forward the datagram to the correct router.
  - However, to update the routing table of the host, it sends a redirection message to the host

Query Messages



- The echo-request and echo-reply messages can be used to determine if there is active communication at the IP level and obtain statistical information
- Timestamp Request and Reply can be used to determine the round-trip time needed for an IP datagram to travel between them
- To obtain its mask, a host sends an address-mask-request message to a router on the LAN
- A host can broadcast (or multicast) a router-solicitation message. The router(s) that receive the solicitation message broadcast their routing information using the router-advertisement message

Self Study

- Internet Group Management Protocol (IGMP)

Next : Other N/W Layer Protocols and Routing